

## **Project Two README**

Developer: Elijah Bastien

Company: Global Rain Software Engineering

Client: Grazioso Salvare

### **About the Project**

This project is the final full stack delivery developed for Grazioso Salvare, an innovative international rescue animal training company that identifies shelter dogs for search and rescue operations. Building upon the custom MongoDB database layer and CRUD Python module established in Project One, this application introduces an interactive, web based client/server dashboard application.

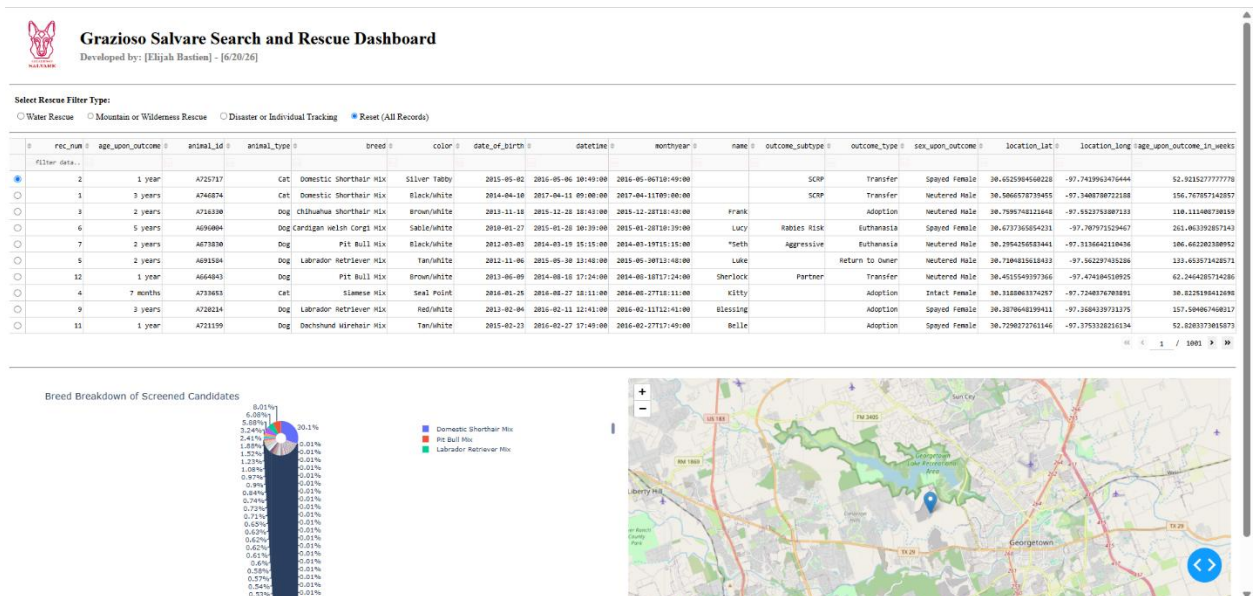
The primary business goal is to enable users at Grazioso Salvare to easily filter, explore, and analyze a large regional dataset of shelter animals provided by a non profit agency near Austin, Texas. The software streamlines the identification of candidate dogs matching specific physical and breed profiles tailored to distinct rescue disciplines.

### **Functional Proof**

Here is the executions demonstrating the fully functional user interface, including the Grazioso Salvare company logo, developer name identifier, interactive data table with pagination/sorting, a dynamic breed distribution chart, and an interactive geolocation map:

## 1. Initial / Reset State

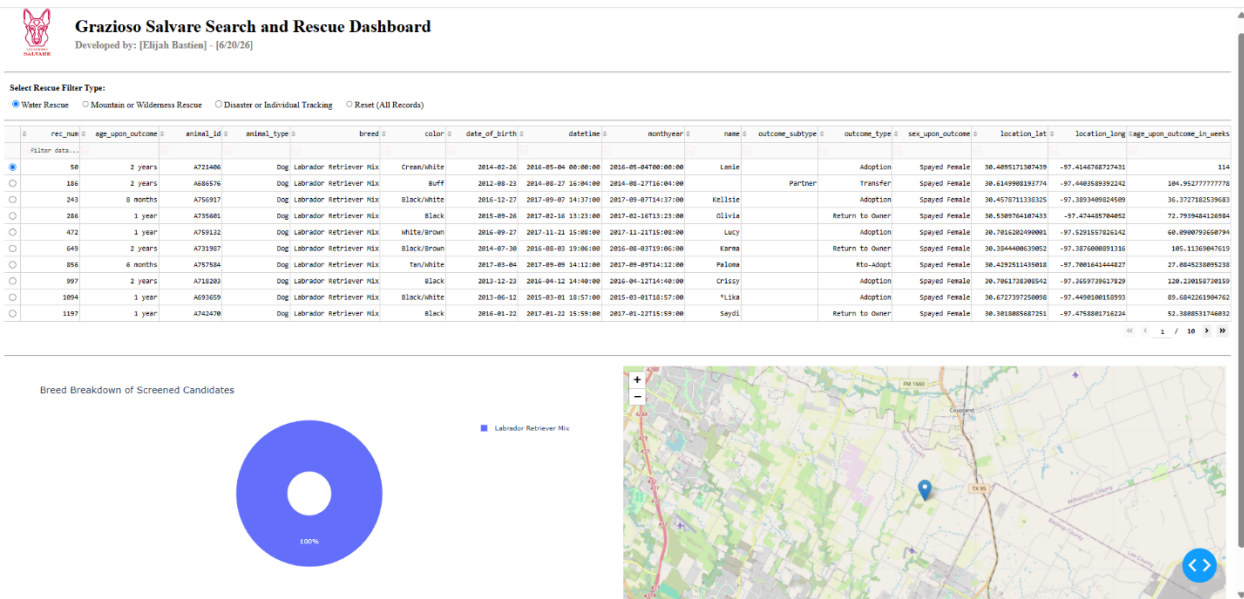
- Filter Selection: "Reset (All Records)"
- Description: Displays the baseline state of the dashboard upon initial server startup or after clearing active queries. The data table displays the full, unfiltered dataset via a retrieve all query, and the map defaults to the initial indexed record.



## 2. Water Rescue Profile

- Filter Selection: "Water Rescue"
- Description: Dynamically updates the data table to display only dogs matching the Water Rescue profile (Labrador Retriever Mix, Chesapeake Bay Retriever, or Newfoundland;

Spayed Female; aged 26 to 156 weeks). The pie chart and map scale smoothly to reflect only these matching candidates.



### 3. Mountain or Wilderness Rescue Profile

- Filter Selection: "Mountain or Wilderness Rescue"
- Description: Dynamically updates all widgets to isolate dogs fitting the Mountain Rescue profile (German Shepherd, Alaskan Malamute, Siberian Husky, Rottweiler, or St. Bernard; Intact Male; aged 26 to 156 weeks).



## Grazioso Salvare Search and Rescue Dashboard

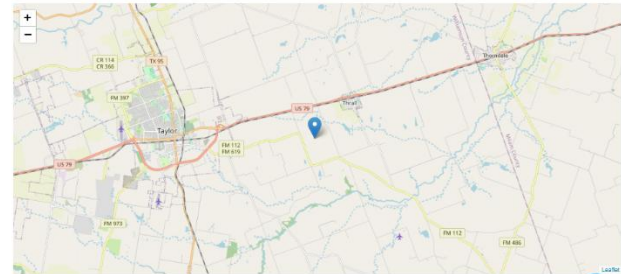
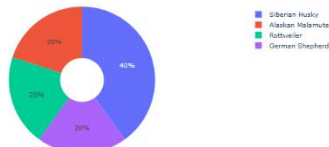
Developed by: [Elijah Bastien] - [6/20/26]

### Select Rescue Filter Type:

☐ Water Rescue ☒ Mountain or Wilderness Rescue ☐ Disaster or Individual Tracking ☐ Reset (All Records)

| s                                | rec_num | age_upon_outcome | animal_id | animal_type | breed            | color       | date_of_birth | datetime            | monthyear           | name    | outcome_subtype | outcome_type    | sex_upon_outcome | location_lat     | location_long     | age_upon_outcome_in_weeks |
|----------------------------------|---------|------------------|-----------|-------------|------------------|-------------|---------------|---------------------|---------------------|---------|-----------------|-----------------|------------------|------------------|-------------------|---------------------------|
| Filter data...                   | 3130    | 2 years          | A721834   | Dog         | Siberian Husky   | Brown/White | 2014-03-05    | 2016-03-12 16:23:00 | 2016-03-23T16:23:00 |         | Suffering       | Euthanasia      | Intact Male      | 38.568999440899  | -97.328559408025  | 107.89751994127           |
| <input checked="" type="radio"/> | 5115    | 2 years          | A708726   | Dog         | Alaskan Malamute | Sable/White | 2013-07-30    | 2015-06-02 17:24:00 | 2015-06-07T17:24:00 | Papa    |                 | Return to Owner | Intact Male      | 38.438933591938  | -97.408825367937  | 104.817857142857          |
| <input type="radio"/>            | 6023    | 2 years          | A728345   | Dog         | Rottweiler       | Black       | 2015-09-31    | 2017-09-23 11:23:00 | 2017-09-23T11:23:00 | Zake    |                 | Return to Owner | Intact Male      | 38.46657286743   | -97.557352893826  | 128.924988793651          |
| <input type="radio"/>            | 6193    | 2 years          | A704381   | Dog         | Siberian Husky   | Black/White | 2013-06-01    | 2015-06-02 16:41:00 | 2015-06-02T16:41:00 | LOBO    |                 | Return to Owner | Intact Male      | 38.4263764229275 | -97.4389581796886 | 104.522767094127          |
| <input type="radio"/>            | 6557    | 6 months         | A765461   | Dog         | German Shepherd  | Sable       | 2017-07-28    | 2018-01-22 11:54:00 | 2018-01-22T11:54:00 | Sargent |                 | Return to Owner | Intact Male      | 38.486669958851  | -97.485688934264  | 26.642261967619           |

Breed Breakdown of Screened Candidates



## 4. Disaster or Individual Tracking Profile

- Filter Selection: "Disaster or Individual Tracking"
- Description: Dynamically updates all widgets to isolate dogs matching the Disaster Tracking profile (Doberman Pinscher, Golden Retriever, Bloodhound, or Rottweiler; Intact Male; aged 20 to 300 weeks).

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## Grazioso Salvare Search and Rescue Dashboard

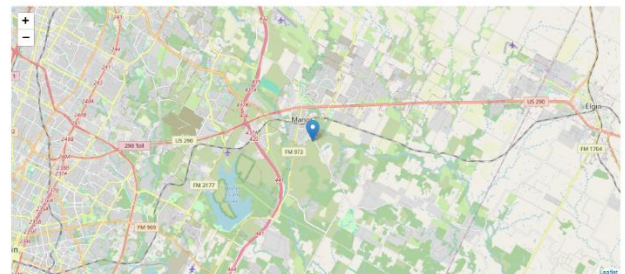
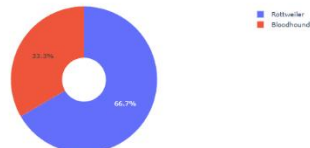
Developed by: [Elijah Bastien] - [6/20/26]

### Select Rescue Filter Type:

☐ Water Rescue ☐ Mountain or Wilderness Rescue ☒ Disaster or Individual Tracking ☐ Reset (All Records)

| s                     | rec_num | age_upon_outcome | animal_id | animal_type | breed      | color       | date_of_birth | datetime            | monthyear           | name    | outcome_subtype | outcome_type    | sex_upon_outcome | location_lat     | location_long     | age_upon_outcome_in_weeks |
|-----------------------|---------|------------------|-----------|-------------|------------|-------------|---------------|---------------------|---------------------|---------|-----------------|-----------------|------------------|------------------|-------------------|---------------------------|
| Filter data...        | 2957    | 4 years          | A094614   | Dog         | Rottweiler | Black/Brown | 2011-01-01    | 2015-01-01 14:25:00 | 2015-01-01T14:25:00 | Striker |                 | Return to Owner | Intact Male      | 38.325873285611  | -97.540296618502  | 206.808891026349          |
| <input type="radio"/> | 3767    | 4 years          | A732291   | Dog         | Bloodhound | Red         | 2011-09-28    | 2015-09-22 15:43:00 | 2015-09-22T15:43:00 | Boomer  |                 | Return to Owner | Intact Male      | 38.2789983763287 | -97.5823916912722 | 209.89381837882           |
| <input type="radio"/> | 6021    | 2 years          | A728345   | Dog         | Rottweiler | Black       | 2015-09-31    | 2017-09-23 11:23:00 | 2017-09-23T11:23:00 | Zake    |                 | Return to Owner | Intact Male      | 38.46657286743   | -97.557352893826  | 128.924988793651          |

Breed Breakdown of Screened Candidates



## **Technical Decisions**

MongoDB: was used as the data repository because its document oriented scheme aligns well with variable shelter tracking records. If a structured SQL platform was used unstructured fields or missing attributes within animal records would disrupt data queries.

The application communicates with the database layer via the official PyMongo library driver. PyMongo maps MongoDB BSON documents natively into standard Python dictionaries and lists, eliminating ORM middleware layers.

The web platform is structured using Dash which provides the infrastructure for implementing both the View and Controller completely in Python.

Dash elements ( ex. `dccc.graph`) act as web render visual widgets, providing highly responsive and intuitive interface for non technical administrators.

Dash callbacks function as the controller middleware. It links reactive inputs from interactive dashboard buttons to specific logic routines, the dashboard coordinates background MongoDB execution queries through the `AnimalShelter` class. Then it forces outputs to re render the data table, data visualizations, and geographic marker layouts on the fly without refreshing the web browser.

## **Steps for Project Replication**

Verify Database Middleware: Ensure your custom `CRUD_Python_Module.py` file containing the `AnimalShelter` class is located inside the root project execution directory.

Launch Database Instance: Initiate your local or networked MongoDB daemon service and ensure the aac database context and its nested animals records array collection are fully accessible.

Install Core Dependencies: Ensure the target execution space has all structural libraries properly installed via pip:

```
pip install jupyter-dash dash-leaflet dash plotly pandas numpy matplotlib
```

Configure Local Images: Place the high-resolution company branding logo (GraziosoSalvareLogo.png) within the active workspace folder.

Execute Codebase: Open the dashboard file (ProjectTwoDashboard.ipynb) within your Jupyter environment and click Run All Cells. Click the automatically configured server address port link output at the bottom of the script window to launch the interactive UI application.

## **Challenges**

When fetching records directly from MongoDB via PyMongo into a pandas DataFrame, the internal `_id` field returned an invalid JSON object serialization type (ObjectID), which broke the Dash data table layout rendering routine.

Resolution: Solved by writing programmatic conditional verification drops (if `'_id'` in `df.columns`: `df.drop(columns=['_id'], inplace=True)`) directly into the data preparation phase of the callback routines.